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Teaching AI security at university

A 14-week, 3-credit module in AI security for fourth-year or master's students in computer science, data science or cybersecurity. Prerequisites: machine learning and basic security. The four Darsi lectures anchor weeks 3, 6, 9 and 11. All attacks are practised only on sandboxed systems under written rules of engagement.

Learning outcomes

1. State threat models for machine-learning systems.
2. Generate adversarial examples and evaluate defences with adaptive attacks.
3. Analyse poisoning and backdoor risks in data supply chains.
4. Measure privacy leakage and apply differential privacy.
5. Secure LLM applications and conduct scoped red teaming.

14-week syllabus

Week	Topic	In class and out
1	Threat models for ML systems	Rules of engagement signed
2	ML refresher: losses and gradients	Problem set 1
3	Adversarial examples and FGSM	Lab 1: FGSM and PGD
4	Adversarial training and robust evaluation	Attack-the-defence assignment
5	Physical and black-box attacks	Problem set 2
6	Data poisoning and backdoors	Lab 2: plant and detect
7	Data supply chain and provenance	Midterm exam
8	Model extraction and inversion	Problem set 3
9	Membership inference and differential privacy	Lab 3: attack and DP-SGD
10	Federated learning and privacy in practice	Privacy memo
11	Prompt injection and LLM applications	Lab 4: scoped red team
12	Securing agents and tool use	Project start
13	Governance: NIST AI RMF and incident response	Project checkpoint
14	Project presentations and review	Final exam

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Lecture slide outline

Lecture 1: Adversarial examples

- Stickers on a stop sign
- Why linearity makes models fragile
- FGSM and PGD
- Adversarial training and adaptive evaluation
- Activity: attack, defend, attack again

Lecture 2: Poisoning and backdoors

- A chatbot offline in a day
- Availability, targeted, backdoor
- Why clean accuracy hides backdoors
- Provenance and filtering
- Activity: map the supply chain

Lecture 3: Privacy attacks

- Anonymous ratings, re-identified
- Membership inference
- Differential privacy and the Laplace mechanism
- DP-SGD and the privacy budget
- Activity: randomised response

Lecture 4: Prompt injection

- An email that gave orders
- Direct and indirect injection
- Why filters fail
- Least privilege, approval, logging
- Activity: red team in scope

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Problem set

1. A linear classifier has $w = (0.5, -3, 1, 2)$ and score $w^{T}x + b = 1.2$ (positive class). What is the smallest L^{∞} budget ε for which FGSM can flip the prediction?

2. A paper reports 70% robust accuracy against FGSM at $\varepsilon = 8/255$ and no other attacks. List three weaknesses in this evaluation.

3. A backdoor is planted in 0.5% of 40,000 training images. Clean accuracy is 97% and 94% of triggered test images are misclassified to the target. How many images were poisoned, and which number shows the attack?

4. A university releases three counts, each with Laplace noise of scale 4. Each count has sensitivity 1. What ε does each release use, and what is the total under basic composition?

5. A model reaches 100% training accuracy and 72% test accuracy. Explain why it is vulnerable to membership inference and name two mitigations, one with a formal guarantee.

6. An LLM assistant can read a lecturer's inbox and send email. Give a threat model for indirect prompt injection and three design controls.

Solutions

> **1.** The score can drop by $\varepsilon\|w\|_1 = 6.5\varepsilon$. Flipping needs $6.5\varepsilon > 1.2$, so $\varepsilon > 0.185$ (about 0.19).

> **2.** No iterative attack such as PGD; no adaptive attack designed for the defence; no check for gradient masking (for example, black-box attacks outperforming white-box ones, or accuracy not reaching zero as ε grows).

> **3.** 200 images. The 94% attack success rate shows the attack; 97% clean accuracy hides it.

> **4.** Scale = Δ/ε, so $\varepsilon = 1/4 = 0.25$ each; the total is $3 \times 0.25 = 0.75$.

> **5.** The large gap means it behaves differently on members, which an attacker can detect from confidence. Regularisation or early stopping reduces leakage; DP-SGD gives a formal bound.

> **6.** An outsider emails hidden instructions that the assistant reads, aiming to exfiltrate mail. Controls: no send without human approval showing recipients; restrict sending to known addresses; treat model output as untrusted and log all tool calls.

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Lab handout: Lab 4: Red team in scope

Goal

Attack a sandboxed LLM course assistant within written rules, then mitigate and re-test.

Steps

1. Read and sign the scope and rules of engagement.
2. Attempt direct injection and log each attempt.
3. Plant indirect injections in documents the assistant retrieves.
4. Implement one mitigation: least privilege, approval or output handling.
5. Re-run all attempts and report success rates before and after.

Hand in

A findings report using the template, with before-and-after rates.

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Assessment rubric

Criterion	Excellent	Good	Developing	Not yet
Threat model	Clear, justified and scoped	Clear goal and capability	Vague	Missing
Attack and evaluation	Adaptive, with rates and uncertainty	Adaptive attacks	Standard attacks only	Weak or untested
Mitigation	Implemented, re-tested and its limits explained	Implemented and re-tested	Proposed only	None
Ethics and reporting	Clear, reproducible, responsible	Clear report	In scope, poor report	Out of scope

Reading list

Goodfellow, I. J., Shlens, J., & Szegedy, C. (2015). Explaining and harnessing adversarial examples. ICLR.
FGSM and the linearity hypothesis.

Madry, A., Makelov, A., Schmidt, L., Tsipras, D., & Vladu, A. (2018). Towards deep learning models resistant to adversarial attacks. ICLR.
PGD and adversarial training.

Carlini, N., Athalye, A., Papernot, N., Brendel, W., Rauber, J., Tsipras, D., Goodfellow, I., Madry, A., & Kurakin, A. (2019). On evaluating adversarial robustness. arXiv:1902.06705.
How to evaluate defences.

Gu, T., Dolan-Gavitt, B., & Garg, S. (2017). BadNets: Identifying vulnerabilities in the machine learning model supply chain. arXiv:1708.06733.
Backdoor attacks.

Shokri, R., Stronati, M., Song, C., & Shmatikov, V. (2017). Membership inference attacks against machine learning models. IEEE Symposium on Security and Privacy.
Membership inference.

Dwork, C., & Roth, A. (2014). The algorithmic foundations of differential privacy. Foundations and Trends in Theoretical Computer Science, 9(3–4), 211–407.
Differential privacy.

Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., & Zhang, L. (2016). Deep learning with differential privacy. ACM CCS.
DP-SGD.

Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Holz, T., & Fritz, M. (2023). Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. ACM AISec Workshop.
Indirect prompt injection.

Vassilev, A., Oprea, A., Fordyce, A., & Anderson, H. (2024). Adversarial machine learning: A taxonomy and terminology of attacks and mitigations (NIST AI 100-2 E2023). NIST.
Attack taxonomy.

OWASP Foundation. OWASP Top 10 for Large Language Model Applications. https://genai.owasp.org/ LLM application risks.